

Work Summary Report

Customer Management to Loan Module – Microfinance Application (Samitha Urban Enterprise Project)

This document summarises the work completed across the core banking modules from Customer Management through the Loan Module of the microfinance application. It is organised in two parts: first, the business functionality worked on and how it fits into the overall system; second, the technical techniques used to build it. Wherever a feature already existed in the codebase, contributions are described as enhancements rather than as new work.

Part 1 – Business Functionality

A. Customer & Member Onboarding Lifecycle

When a new individual joins the institution, the system registers their personal, demographic, contact, and nominee information, generates a unique Member Code, collects KYC identity records, and establishes their foundational account in the core banking database.

My contribution: I worked on the end-to-end registration flow – implementing multi-document KYC verification (Aadhaar, PAN, Voter ID, Ration Card), photo and signature upload handling with live image previews, automatic default Savings Account provisioning with zero opening friction upon member creation, Aadhaar visibility toggle, and standardized uppercase normalization across stored customer records.

B. Share Capital & Distinctive Share Management

As a cooperative/Nidhi financial structure, every member must hold capital shares. The shareholding module manages share allocation, tracks unallotted equity inventory, records share transfers between members, and issues legal share certificates.

My contribution: I implemented the share transfer transaction logic ensuring share ledger balance consistency between transferor and transferee, engineered the Distinctive Number Order (DNO) allocation and automated regeneration engine, and developed the formal share certificate generation layout with distinct certificate numbering.

C. General Ledger & Double-Entry Accounting Integration

The accounting backbone provides institutional double-entry bookkeeping across all branch operations – capturing cash/bank receipt and payment vouchers, contra transfers, cheque clearance lifecycles, and real-time statutory financial statements.

My contribution: I connected cash desk and teller operations directly to the double-entry ledger so that every deposit, withdrawal, loan disbursement, and EMI repayment automatically posts balanced debit and credit entries. I also verified the automated financial reporting logic for the Daybook, Cashbook, Trial Balance, Profit & Loss Statement, and Balance Sheet.

D. Customer Savings & Deposit Operations

This module serves as the primary transaction hub for customer liquidity – offering savings accounts, cash desk deposits and withdrawals, account-to-account internal fund transfers, passbook record generation, and automated interest crediting.

My contribution: I built the cash teller deposit/withdrawal validation, eliminated duplicate alert popups on passbook transactions, engineered the automated quarterly interest computation engine with a single-click batch transfer button, and developed the customer savings statement view featuring date-range filters, summary metric cards, and instant PDF download/print capabilities.

Comparison of Core Financial Products across the Pipeline

Scheme / Product	How Money is Deposited / Borrowed	How Interest is Worked Out	What the Customer Receives / Pays
Savings Account	Anytime flexible deposits and cash teller withdrawals	Quarterly interest credited to savings balance based on daily balance	Liquidity on demand, quarterly interest yield, passbook records
RD (Recurring Deposit)	Fixed monthly installments over chosen term	Compounded monthly or quarterly over term	Total saved sum plus accumulated interest paid at maturity
FD (Fixed Deposit)	Single one-time lump sum deposit	Calculated quarterly/yearly depending on chosen scheme	Lump sum principal plus compounded interest at maturity
MIS (Monthly Income)	Single lump sum deposit kept locked in	Fixed monthly interest calculated on principal	Monthly interest payout to Savings Account; principal returned at maturity
DRD (Daily Deposit)	Small daily cash collections via field collectors	Calculated daily/monthly over specified number of days	Total saved capital plus interest returned at maturity
Regular / Term Loans	Disbursed lump sum net of processing & statutory deductions	Reducing balance or flat rate EMI schedule	Repays monthly EMIs; receives NOC upon full clearance

E. Scheme & Policy Management (RD, FD, DRD, MIS)

The investment module enables long-term deposits across Recurring Deposits (RD), Fixed Deposits (FD), Daily Recurring Deposits (DRD), and Monthly Income Schemes (MIS) â€” tracking policy booking, installment schedules, renewal payments, and maturity payouts.

My contribution: I worked on plan interest and maturity calculations across all four deposit types, connected policy creation to savings account balance checks, built the rapid daily renewal collection screens for field collectors, and added installment record book tracking with automated late-fee/penalty calculations.

F. Credit & Loan Management Lifecycle (Core Asset Engine)

This constitutes the primary asset management engine of the institution: from initial loan eligibility assessment and dynamic EMI calculation to application appraisal, multi-tier sanctioning, deduction schedule execution, disbursement, monthly installment collections, legal documentation, and loan foreclosure.

1. Loan Applied Eligibility & appraisal	2. Verification Co-applicant & guarantor	3. Sanctioned Authority approval	4. Disbursed Fee deductions & net payout	5. EMI Repayment Monthly installment collections	6. Closed / NOC Full clearance & certificate
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My contribution: I built the dynamic loan deduction engine (automatically computing processing fees, insurance premiums, legal fees, and GST dynamically per loan product at disbursement), integrated the automated notification service for SMS/email alerts on sanction and disbursement, developed the regular loan statement ledger, and implemented OpenPDF document printing for statutory loan agreements, promissory notes, and NOC clearance certificates.

G. Extended Lending (Secured Gold & Joint Liability Loans)

To support diverse borrower segments, the loan ecosystem extends into Secured Jewellery / Gold Loans (with ornament appraisal, purity karat valuation, and LTV enforcement) and Joint Liability Group (JLG) loans (group formation and peer-guaranteed lending).

My contribution: I connected gold ornament appraisal logic to maximum permissible sanction limits, implemented gold loan closure with collateral packet release tracking, and wired lending group directories with peer-guarantee verification.

How These Pieces Fit Together

At the architectural center of the application sits the Customer/Member record. Every share certificate, savings account, deposit policy (RD/FD/DRD/MIS), and loan account is tied to this central Member ID. When a member applies for a loan or policy, the system verifies their KYC and linked Savings Account. Loan disbursements can credit their savings account directly, and EMI repayments or policy renewals can automatically deduct from their available balance. Simultaneously, every financial event posts balanced vouchers into the General Ledger, ensuring real-time reconciliation between branch tellers,

portfolio ledgers, and financial balance sheets.

Part 2 – Technical Implementation

The table below lists the primary software engineering techniques, architectural patterns, and libraries utilised across these modules, following the structured pattern: "I used X to achieve Y in the Z business flow."

Technique	How I used it, and why
@Transactional	I used this to make sure that money-related operations (creating a policy, deducting or crediting a Savings Account, running quarterly interest batches, and disbursing loans) either complete fully or not at all – so a failure partway through never leaves a balance or ledger in an inconsistent state.
@Scheduled / Batch Processing	I used this to automate recurring calculations – the system checks and processes quarterly savings interest crediting and scheduled policy updates without requiring manual row-by-row intervention.
Dynamic Fee Deduction Engine	I used this to dynamically compute processing charges, insurance premiums, legal fees, and GST on loan disbursement, so deductions are auto-calculated from scheme rules rather than hardcoded.
Custom Business Exceptions & Validations	I used these to enforce business rules cleanly – for example, blocking savings withdrawals below minimum balance, preventing duplicate passbook alerts, and validating guarantor eligibility.
DTOs (Data Transfer Objects)	I used these to validate and structure the data coming in from customer registration, loan origination, and policy forms before saving, catching missing or malformed inputs early.
PDF Generation (OpenPDF & Flying Saucer)	I built the downloadable document generation features using these libraries – staff can now download formatted Customer Statements, Share Certificates, Loan Agreements, and NOCs on demand.
Spreadsheet Integration (Apache POI)	I used this to generate downloadable Excel reports for member registers and financial audits with proper column typing and date formatting.
HandlerInterceptor & Activity Logging	I used this to log staff actions across request routes, recording timestamps and mutation details into the ActivityLog table to maintain an audit trail for compliance.
AJAX & Dynamic DOM Rendering	I used this across customer search, loan statement filtering, and passbook views to fetch ledger entries asynchronously and update summary cards without reloading the full page.

Smaller Updates & Polish Across These Modules

- Added new customer form fields (occupation, education, monthly income, member fees, building fund, admin charges).
- Standardised all personal customer text fields to a consistent upper-case format upon database persistence.
- Resolved account number binding bug in Customer Summary and Savings Record Book views.
- Implemented Aadhaar visibility toggle and enhanced document preview for KYC verification.
- Eliminated duplicate alert popups on passbook transaction button clicks.
- Integrated dynamic loan deduction details (processing fees, insurance, legal charges, GST) calculated at disbursement.
- Added automated loan notification service hooks dispatching SMS/email on status changes.
- Applied unified Samitha Urban corporate branding, color schemes, and responsive UI styling.

Summary

The core pipeline from Customer Management to Loan Management represents the complete foundational banking cycle of the Samitha Urban Microfinance system. Major deliverables include verified member onboarding with instant savings account provisioning, distinctive shareholding management, double-entry general ledger integration, multi-scheme deposit lifecycles with batch interest crediting, and a full-featured loan management system complete with dynamic deduction schedules, EMI amortization, OpenPDF legal document printing, and foreclosure settlement. The implementation adheres to enterprise Java/Spring Boot standards, ensuring robust transactional consistency, clean modular separation, and readiness for client demonstration and production operations.